

$$\begin{aligned}
& \rightarrow \frac{1}{16\pi^2} \left(m_1 \tilde{\mu} g_1^4 (-C_{\phi 1\phi 2} + m_1 \tilde{\mu}) LF_{2,2,0}[m_1, \tilde{\mu}] + \right. \\
& \frac{1}{3} m_2 \tilde{\mu} g_2^4 (-5 C_{\phi 1\phi 2} + 9 m_2 \tilde{\mu}) LF_{2,2,0}[m_2, \tilde{\mu}] - \frac{2}{3} m_2 C_{\phi 1\phi 2} \tilde{\mu} g_2^4 LF_{3,2,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{2} C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_e^{pr}} a_e^{pr} LF_{4,1,-1}[m_l^p, m_e^r] - \frac{2}{3} C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_e^{pr}} a_e^{pr} LF_{5,1,-2}[m_l^p, m_e^r] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_d^{pr}} a_d^{pr} LF_{4,1,-1}[m_q^p, m_d^r] - 2 C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_d^{pr}} a_d^{pr} LF_{5,1,-2}[m_q^p, m_d^r] + \\
& C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{2,2,0}[m_q^p, m_u^r] - \frac{1}{2} C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,2,-1}[m_q^p, m_u^r] - \\
& C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,1,0}[m_u^r, m_q^p] - C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,2,-1}[m_u^r, m_q^p] + \\
& 3 C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{4,1,-1}[m_u^r, m_q^p] - 2 C_{\phi 1\phi 2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{5,1,-2}[m_u^r, m_q^p] - \\
& \frac{5}{2} m_1 C_{\phi 1\phi 2} \tilde{\mu} g_1^4 LF_{3,2,-1}[\tilde{\mu}, m_1] + g_1^4 m_1^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,2,0}[\tilde{\mu}, m_1] + \\
& \frac{1}{2} m_1 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 LF_{4,1,-1}[\tilde{\mu}, m_1] + 2 m_1 C_{\phi 1\phi 2} \tilde{\mu} g_1^4 LF_{4,2,-2}[\tilde{\mu}, m_1] - \\
& g_1^4 m_1^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,2,-1}[\tilde{\mu}, m_1] - \frac{2}{3} m_1 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 LF_{5,1,-2}[\tilde{\mu}, m_1] - \\
& \frac{4}{3} m_2 C_{\phi 1\phi 2} \tilde{\mu} g_2^4 LF_{3,1,0}[\tilde{\mu}, m_2] - \frac{59}{6} m_2 C_{\phi 1\phi 2} \tilde{\mu} g_2^4 LF_{3,2,-1}[\tilde{\mu}, m_2] + \\
& 3 g_2^4 m_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,2,0}[\tilde{\mu}, m_2] + \frac{7}{2} m_2 C_{\phi 1\phi 2} \tilde{\mu} g_2^4 LF_{4,1,-1}[\tilde{\mu}, m_2] + \\
& 8 m_2 C_{\phi 1\phi 2} \tilde{\mu} g_2^4 LF_{4,2,-2}[\tilde{\mu}, m_2] - 4 g_2^4 m_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,2,-1}[\tilde{\mu}, m_2] - \\
& 2 m_2 C_{\phi 1\phi 2} \tilde{\mu} g_2^4 LF_{5,1,-2}[\tilde{\mu}, m_2] - 6 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& 4 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{2,2,1,0}[m_2, \tilde{\mu}, m_1] + 4 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \\
& LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - 4 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& \frac{1}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{sr} a_e^{pt} LF_{2,2,1,-1}[m_l^p, m_l^s, m_e^t] + \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_d^{pr}} \overline{y_u^{st}} y_u^{pt} a_d^{sr} \\
& LF_{2,2,1,-1}[m_q^p, m_q^s, m_d^r] - \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{sr} a_d^{pt} LF_{2,2,1,-1}[m_q^p, m_q^s, m_d^t] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{y_d^{pr}} y_d^{sr} - \overline{y_u^{pr}} y_u^{sr}) LF_{2,1,1,0}[m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (-\overline{y_d^{pr}} y_d^{sr} + \overline{y_u^{pr}} y_u^{sr}) LF_{3,1,1,-1}[m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{4} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (-\overline{y_d^{pr}} y_d^{sr} + \overline{y_u^{pr}} y_u^{sr}) LF_{2,2,1,-1}[m_q^p, m_u^t, m_q^s] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{y_d^{pr}} y_d^{sr} - \overline{y_u^{pr}} y_u^{sr}) LF_{2,1,1,0}[m_q^s, m_q^p, m_u^t] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (-\overline{y_d^{pr}} y_d^{sr} + \overline{y_u^{pr}} y_u^{sr}) LF_{3,1,1,-1}[m_q^s, m_q^p, m_u^t] + \\
& \frac{3}{4} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (-\overline{y_d^{pr}} y_d^{sr} + \overline{y_u^{pr}} y_u^{sr}) LF_{2,2,1,-1}[m_q^s, m_u^t, m_q^p] + \\
& 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] - C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 (m_1 + 7 m_2) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
& 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,1,1,0}[\tilde{\mu}, m_1, m_2] + 6 m_2 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \\
& 3 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
& 2 C_{\phi 1\phi 2} \tilde{\mu} g_1^2 g_2^2 (m_1 + 3 m_2) LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - 4 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 (C_{\phi 1^2} + C_{\phi 2^2}) \\
& LF_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] - 3 \tilde{\mu}^2 \overline{y_d^{pr}} \overline{y_d^{st}} a_d^{pt} a_d^{sr} LF_{1,1,1,0}[m_d^r, m_d^t, m_q^p, m_q^s] - \\
& \frac{3}{2} \tilde{\mu} \overline{y_d^{st}} a_d^{pt} a_d^{sr} (C_{\phi 1\phi 2} \overline{a_d^{pr}} + \tilde{\mu} \overline{y_d^{pr}} (C_{\phi 1^2} - C_{\phi 2^2})) LF_{2,2,1,1,-1}[m_d^r, m_q^p, m_d^t, m_q^s] - \\
& \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} \overline{y_d^{st}} a_d^{pt} (a_d^{sr} (-C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_d^{sr}) LF_{2,2,1,1,-1}[m_d^r, m_q^s, m_d^t, m_q^p] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_d^{st}} \overline{a_d^{pr}} a_d^{pt} a_d^{sr} LF_{2,2,1,1,-1}[m_d^t, m_q^p, m_d^r, m_q^s] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu}^3 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{sr} a_d^{pt} LF_{2,2,1,1,-1}[m_d^t, m_q^s, m_d^r, m_q^p] - \\
& \tilde{\mu}^2 \overline{y_e^{pr}} \overline{y_e^{st}} a_e^{pt} a_e^{sr} LF_{1,1,1,0}[m_e^r, m_e^t, m_l^p, m_l^s] - \\
& \frac{1}{2} \tilde{\mu} \overline{y_e^{st}} a_e^{pt} a_e^{sr} (C_{\phi 1\phi 2} \overline{a_e^{pr}} + \tilde{\mu} \overline{y_e^{pr}} (C_{\phi 1^2} - C_{\phi 2^2})) LF_{2,2,1,1,-1}[m_e^r, m_l^p, m_e^t, m_l^s] - \\
& \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{pr}} \overline{y_e^{st}} a_e^{pt} (a_e^{sr} (-C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_e^{sr}) LF_{2,2,1,1,-1}[m_e^r, m_l^s, m_e^t, m_l^p] + \\
& \frac{1}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_e^{st}} \overline{a_e^{pr}} a_e^{pt} a_e^{sr} LF_{2,2,1,1,-1}[m_e^t, m_l^p, m_e^r, m_l^s] + \\
& \frac{1}{2} C_{\phi 1\phi 2} \tilde{\mu}^3 \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{sr} a_e^{pt} LF_{2,2,1,1,-1}[m_e^t, m_l^s, m_e^r, m_l^p] - \\
& \tilde{\mu} \overline{y_e^{st}} a_e^{pt} a_e^{sr} (C_{\phi 1\phi 2} \overline{a_e^{pr}} + \tilde{\mu} \overline{y_e^{pr}} (C_{\phi 1^2} + C_{\phi 2^2})) LF_{2,1,1,1,0}[m_l^p, m_e^r, m_e^t, m_l^s] + \\
& \tilde{\mu} \overline{y_e^{st}} a_e^{pt} a_e^{sr} (C_{\phi 1\phi 2} \overline{a_e^{pr}} + \tilde{\mu} \overline{y_e^{pr}} (C_{\phi 1^2} + C_{\phi 2^2})) LF_{3,1,1,1,-1}[m_l^p, m_e^r, m_e^t, m_l^s] - \\
& C_{\phi 1\phi 2} \tilde{\mu}^3 \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{sr} a_e^{pt} LF_{2,1,1,1,0}[m_l^s, m_e^r, m_e^t, m_l^p] + \\
& C_{\phi 1\phi 2} \tilde{\mu}^3 \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{sr} a_e^{pt} LF_{3,1,1,1,-1}[m_l^s, m_e^r, m_e^t, m_l^p] - \\
& 3 \tilde{\mu} \overline{y_d^{st}} a_d^{pt} a_d^{sr} (C_{\phi 1\phi 2} \overline{a_d^{pr}} + \tilde{\mu} \overline{y_d^{pr}} (C_{\phi 1^2} + C_{\phi 2^2})) LF_{2,1,1,1,0}[m_q^p, m_d^r, m_d^t, m_q^s] + \\
& 3 \tilde{\mu} \overline{y_d^{st}} a_d^{pt} a_d^{sr} (C_{\phi 1\phi 2} \overline{a_d^{pr}} + \tilde{\mu} \overline{y_d^{pr}} (C_{\phi 1^2} + C_{\phi 2^2})) LF_{3,1,1,1,-1}[m_q^p, m_d^r, m_d^t, m_q^s] + \frac{3}{2} \tilde{\mu} \overline{y_u^{st}} \\
& (C_{\phi 1\phi 2} \overline{a_d^{pr}} a_d^{sr} a_u^{pt} + \tilde{\mu} \overline{y_d^{pr}} (C_{\phi 1\phi 2} \tilde{\mu} y_d^{sr} a_u^{pt} + a_d^{sr} (2 a_u^{pt} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{pt}))) \\
& LF_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^r, m_u^t] + \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_d^{st}} a_d^{pt} \overline{a_u^{pr}} a_u^{sr} LF_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^t, m_u^r] - \\
& 3 \tilde{\mu}^2 \overline{y_u^{pr}} \overline{y_u^{st}} a_u^{pt} a_u^{sr} LF_{1,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t] - \\
& 3 \tilde{\mu}^2 \overline{y_u^{pr}} \overline{y_u^{st}} a_u^{pt} a_u^{sr} (C_{\phi 1^2} + C_{\phi 2^2}) LF_{2,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t] + \\
& 3 \tilde{\mu}^2 \overline{y_u^{pr}} \overline{y_u^{st}} a_u^{pt} a_u^{sr} (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,1,1,1,-1}[m_q^p, m_q^s, m_u^r, m_u^t] - \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{a_u^{pr}} a_u^{sr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{sr}) LF_{2,1,1,1,0}[m_q^p, m_q^s, m_u^t, m_u^r] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{a_u^{pr}} a_u^{sr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{sr}) LF_{3,1,1,1,-1}[m_q^p, m_q^s, m_u^t, m_u^r] + \frac{3}{4} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} \\
& (-C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{sr} + \tilde{\mu} \overline{y_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} + 3 C_{\phi 1\phi 2} \tilde{\mu} y_u^{sr})) LF_{2,2,1,1,-1}[m_q^p, m_u^r, m_q^s, m_u^t] + \\
& \frac{3}{4} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{sr} + \tilde{\mu} \overline{y_u^{pr}} (4 C_{\phi 2^2} a_u^{sr} + C_{\phi 1\phi 2} \tilde{\mu} y_u^{sr})) \\
& LF_{2,2,1,1,-1}[m_q^p, m_u^t, m_q^s, m_u^r] - 3 C_{\phi 1\phi 2} \tilde{\mu}^3 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{sr} a_d^{pt} LF_{2,1,1,1,0}[m_q^s, m_d^r, m_d^t, m_q^p] + \\
& 3 C_{\phi 1\phi 2} \tilde{\mu}^3 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{sr} a_d^{pt} LF_{3,1,1,1,-1}[m_q^s, m_d^r, m_d^t, m_q^p] - \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{a_u^{pr}} a_u^{sr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{sr}) LF_{2,1,1,1,0}[m_q^s, m_q^p, m_u^t, m_u^r] + \\
& \frac{3}{2} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{a_u^{pr}} a_u^{sr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{sr}) LF_{3,1,1,1,-1}[m_q^s, m_q^p, m_u^t, m_u^r] - \\
& \frac{3}{4} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (-3 \overline{a_u^{pr}} a_u^{sr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{sr}) LF_{2,2,1,1,-1}[m_q^s, m_u^r, m_q^p, m_u^t] + \\
& \frac{3}{4} C_{\phi 1\phi 2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (\overline{a_u^{pr}} a_u^{sr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{sr}) LF_{2,2,1,1,-1}[m_q^s, m_u^t, m_q^p, m_u^r] - 3 \tilde{\mu} \overline{y_u^{st}} a_u^{pt} \\
& (C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{sr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{sr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{sr})) LF_{2,1,1,1,0}[m_u^r, m_q^p, m_q^s, m_u^t] + \\
& 3 \tilde{\mu} \overline{y_u^{st}} a_u^{pt} (C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{sr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{sr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{sr})) \\
& LF_{3,1,1,1,-1}[m_u^r, m_q^p, m_q^s, m_u^t] + \frac{3}{2} \tilde{\mu} \overline{y_u^{st}} a_u^{pt} \\
& (C_{\phi 1\phi 2} \overline{a_u^{pr}} a_u^{sr} + \tilde{\mu} \overline{y_u^{pr}} (a_u^{sr} (C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{sr})) LF_{2,2,1,1,-1}[m_u^t, m_u^r, m_q^p, m_q^s] \Big)
\end{aligned}$$